Pramo Jewels

AI Backend Coding Guide

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Field	Value
Project	Pramo Jewels Premium E-Commerce Platform
Purpose	Define backend development standards for AI-generated server-side code, APIs
Audience	Backend Developers, Solution Architects, AI Coding Assistants, DevOps, QA
Version	1.0

AI Backend Development Standards

Category	Guidelines
Architecture	Follow layered architecture and separation of concerns
REST APIs	Consistent endpoints, validation, versioning and error handling
Business Logic	Encapsulate logic in services; keep controllers lightweight
Database	ORM usage, migrations, transactions and indexed queries
Security	RBAC, JWT/session auth, input validation, encryption
Performance	Caching, pagination, async jobs and optimized queries
Logging	Structured logs, correlation IDs and audit events
Testing	Unit, integration and API contract tests

Objectives

Ensure AI-generated backend code is secure, scalable, maintainable and aligned with the approved system architecture.

Coding Standards

Use descriptive naming, dependency injection, modular services, reusable utilities and project-defined folder conventions.

API Guidelines

Validate requests, return standardized responses, implement idempotency where applicable and document endpoints.

Security Requirements

Prevent SQL injection, XSS and CSRF where relevant, never hard-code secrets, enforce authorization and sanitize all inputs.

Performance Guidelines

Optimize database access, avoid N+1 queries, use caching, queues and background processing for long-running tasks.

Documentation

Document APIs, services, database changes, configuration updates and business rules with every implementation.

Integration

Integrates with Authentication, Database, Cache, Payment Gateway, Shipping APIs, CMS, Analytics, Monitoring and CI/CD.

Acceptance Criteria

AI-generated backend code passes code review, security validation, automated tests, performance benchmarks and conforms to project standards.